

Graphical Abstract

Constraint-Consistent Direction and Diagnostic Encoding Loss in an Externally Validated EEG-Drug CVAE

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Highlights

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- Fold-0: 10/10 direction on TUH/OSF/P-ADIC ($r=0.908/0.851/0.918$).
- Five-fold r is fold-sensitive; prior-only also 10/10 at mean $r=0.636$.
- No paired post-dose EEG; endpoint is signature agreement, not empirical PD transfer.
- CAUEEG magnitude null: Dementia vs Normal $p=0.516$, $d=0.00$ ($n=727$).
- Latent probe AUC=0.579 vs $\theta/\alpha=0.675$ and 2185-D=0.699; head OOF=0.593.

Constraint-Consistent Direction and Diagnostic Encoding Loss in an Externally Validated EEG-Drug CVAE

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Abstract

Chemistry-conditioned EEG digital twins for drug-response simulation must show that constrained, model-generated pharmacology remains consistent under external domain shift, and must separately justify any diagnostic use of twin outputs. We trained a pharmacodynamically constrained CVAE on 2185-D resting-state EEG features with fixed ChemBERTa embeddings for Donepezil and Memantine. Because external cohorts lack paired pre/post-drug EEG, the primary endpoint is concordance of simulated band/connectivity signs with the training constrained signature, not empirical post-dose pharmacodynamic change. On the locked fold-0 checkpoint, direction agreed at 10/10 on TUH EEG ($n=200$; $r=0.908$), an OSF AD/HC set ($n=92$; $r=0.851$), and P-ADIC AD/HC recordings ($n=145$; $r=0.918$). Those high continuous correlations are fold-0 specific: five-fold sensitivity showed much lower mean r (0.385/0.371/0.372). Latent magnitude did not diagnose reliably across four escalating external tests, and encoding probes placed latent ROC-AUC = 0.579 below theta/alpha = 0.675 (best nested head OOF AUC = 0.593). A prior-only signed effect reached 10/10 signs with mean $r=0.636$, so directional agreement is partly recoverable from the literature constraint itself; an equal-weight five-fold ensemble ($r=0.642/0.626/0.627$) did not clearly exceed that prior. The locked fold-0 twin therefore supports signature-concordant response simulation at a selected checkpoint, not fold-stable continuous fidelity, empirical post-dose validation, or external diagnostic utility.

Keywords: pharmacodynamically constrained CVAE, EEG digital twin, drug-conditioned simulation, Alzheimer’s disease, external validation, encoding information loss

1. Introduction

Digital twins and chemistry-conditioned generative models are increasingly used to simulate how a named intervention would change a patient’s physiological features. For such models to be scientifically usable, two questions must be answered separately. First, does a constrained, model-generated drug-response **signature** remain consistent outside the training cohort (sign agreement with the training constrained signature, not paired post-dose pharmacodynamic proof)? Second,

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which twin-derived scalars, if any, carry diagnostic signal under the same external conditions? Conflating those questions invites a common failure mode: a model can look clinically useful on in-distribution diagnosis while its constrained pharmacology does not generalize, or the reverse. Conditional variational autoencoders (CVAEs) and related generative models make that risk concrete, because reconstruction quality, perturbation fidelity, and downstream classification need not move together [1, 2, 3, 4]. Existing generative biomedical models can represent perturbation and outcome relationships, but it remains unclear whether imposing literature-constrained drug-response structure preserves the intended response-direction concordance while retaining independent diagnostic information. What is still scarce is an experimental design that measures those quantities on the same EEG-drug twin and localizes information loss when they dissociate.

Alzheimer’s disease (AD) drug response is a natural stress test for that design. Acetylcholinesterase inhibitors (AChEIs) such as Donepezil help only a subset of patients, with reported response rates often in the 20% to 60% range [5], and predictors spanning genetics, pharmacokinetics, comorbidity, and EEG correlates under inconsistent response definitions [6, 7]. EEG slowing, band-power shifts, and connectivity changes are established AD electrophysiology markers [8, 9, 10, 11, 12, 13], and cholinergic intervention is detectable in spectral EEG [14, 15, 16]. Those facts motivate in silico simulation of Donepezil and Memantine EEG effects. They do not license treating every twin latent as an AD biomarker. Parallel EEG digital-twin work that targets diagnosis or prognosis, including Digital Alzheimer’s Disease Diagnosis (DADD) models, shows that mechanism-informed EEG twins can carry clinical signal when diagnosis is the designed endpoint [17, 18, 19, 20, 21]. The open methodological problem is different: when the twin is built for literature-constrained drug-response simulation, how should external validation separate signature-direction concordance from diagnostic magnitude, and how should residual encoding information be quantified relative to feature-level references?

This paper addresses that measurement gap with a pharmacodynamically constrained CVAE digital twin. The model maps 2185-dimensional resting-state EEG features and chemical Bidirectional Encoder Representations from Transformers (ChemBERTa) drug embeddings for Donepezil and Memantine into a latent space for in silico post-drug feature simulation [22, 23, 24]. We treat band-level response-direction concordance with the training signature as the primary endpoint and test it on three independent external EEG cohorts: Temple University Hospital (TUH) clinical EEG [25], an Open Science Framework (OSF) AD/healthy control set, and p-adic quantum potential EEG (P-ADIC) AD/healthy control recordings [26]. We then test latent drug-response magnitude as a separate diagnostic endpoint across four escalating cohorts, adding Chung-Ang University Hospital EEG (CAUEEG) Normal/mild cognitive impairment (MCI)/Dementia strata [27]. Training features use OpenNeuro accession ds004504 [28]. Finally, we probe CAUEEG baseline latent means against a theta/alpha spectral marker and nested classifier heads to locate any diagnosis-mechanism dissociation at encoding rather than at readout. Domain shift across hospital EEG corpora is treated as part of the test, not as an afterthought [29, 30].

The contributions are as follows:

- We show that, for the locked fold-0 constrained checkpoint, drug-response effect-vector correlations with the training signature reach $r = 0.908, 0.851,$ and 0.918 on TUH, OSF, and P-ADIC, with perfect secondary block-level sign agreement (10/10) on each cohort;

continuous r is fold-sensitive across the five training folds.

- We show that the same twin’s latent drug-response magnitude does not provide reliable external diagnosis across four escalating tests: TUH abnormal versus normal is null ($p = 0.272$, $d = -0.13$); OSF AD versus HC is an exploratory positive only with HC $n = 12$ caveats ($p = 0.000278$, $d = 0.44$); P-ADIC AD versus HC is null ($p = 0.968$, $d = -0.07$); CAUEEG Dementia versus Normal is null ($p = 0.516$, $d = 0.00$, $n = 727$).
- We show that CAUEEG inputs still carry clinical signal (2185-D logistic probe ROC-AUC = 0.699; theta/alpha ROC-AUC = 0.675), that baseline latent means retain a weaker above-chance probe (ROC-AUC = 0.579, permutation $p = 0.010$), and that nested stronger heads peak at out-of-fold ROC-AUC = 0.593, so the tested readout functions do not close the gap to the feature references.
- We therefore quantify, for this constrained CVAE architecture, a concrete measurement result: constraint-consistent response structure can be preserved externally at the locked fold-0 checkpoint, while continuous fidelity is fold-sensitive, diagnostic magnitude does not generalize, and residual clinical information at μ_{base} sits below raw-feature and spectral references.

Section 2 describes the twin architecture, pharmacodynamic constraints, datasets, and endpoints. Section 3 reports constraint-consistent response structure on external EEG, then the escalating magnitude-diagnosis investigation, then the encoding dissection. Section 4 discusses these results against prior twin and AChEI-EEG literature and states architectural implications. Section 5 concludes.

The Methods section next specifies the constrained CVAE, the external cohorts, and the three endpoint families used throughout the results.

2. Methods

The Introduction framed two separate questions: whether a constrained twin preserves a constraint-consistent, model-generated drug-response signature under external EEG domain shift, and whether a twin magnitude endpoint can diagnose externally. This section defines the architecture, constraints, training cohort, external datasets, and the three endpoint families used to answer those questions. Paired post-dose EEG is not part of this study design. Exact configuration keys, script names, and file paths used to regenerate every reported artifact are listed in Supplementary Material S1.

2.1. Overview of the digital twin architecture

Building on that dual-endpoint framing, we specify the generative model used for all experiments. The twin is a pharmacodynamically constrained conditional variational autoencoder (CVAE) with multilayer-perceptron encoders, no graph neural network EEG encoder, no cross-attention fusion, and constrained training loss enabled.

Input EEG is a packed 2185-dimensional feature vector for 19 standard 10-20 channels: power spectral density (PSD; $19 \times 20 = 380$), band powers ($19 \times 5 = 95$; delta, theta, alpha, beta, gamma), and upper-triangle coherence and phase-locking value (PLV) connectivity across five bands (1710 features). The EEG encoder maps structured tensors (PSD, band powers, coherence, PLV) to a 128-dimensional EEG latent. The drug encoder maps a 384-dimensional chemical Bidirectional Encoder Representations from Transformers (ChemBERTa) embedding to a 64-dimensional drug latent. With only two study drugs (Donepezil, Memantine), ChemBERTa supplies a fixed chemical identifier for conditioning rather than a large-library chemical generalization test. Fusion concatenates EEG latent, drug latent, and a scalar disease label (AD = 1, healthy control (HC) = 0) into a 256-dimensional fused representation. The CVAE encoder outputs a Gaussian with mean μ and log-variance over a 128-dimensional latent; the decoder reconstructs the 2185-dimensional feature vector.

At inference, baseline conditioning uses a zero drug vector of length 384. Donepezil and Memantine use their ChemBERTa embeddings. The quantity μ_{base} denotes the CVAE latent mean under zero-drug conditioning for a subject’s EEG. Drug-conditioned means μ_{don} and μ_{mem} use the corresponding drug embeddings with the same EEG encoding path. The architecture data flow (EEG tensors and ChemBERTa drug vector into encoders, fusion with disease label, CVAE μ /log-variance, and feature reconstruction) is illustrated in Fig. 1.

Hidden widths are EEG encoder [256, 512], drug encoder [256, 128], and CVAE [512, 256]. Fusion attention is disabled.

2.2. Pharmacodynamic constraints

Section 2.1 defined where drugs enter the latent path. This subsection states how pharmacodynamic priors shape training, as a deliberate design choice rather than a post hoc regularizer.

Signed EEG effect priors for Donepezil and Memantine are taken from literature-attributed band and connectivity targets. Code comments attribute Donepezil band priors to Babiloni et al. (2006) and Jeong (2004), and Memantine priors to Winblad et al. (2007) and Rive et al. (2013) [16, 8, 23, 24]. The numeric priors used in the loss are:

- Donepezil: delta -0.15, theta -0.20, alpha +0.25, beta +0.10, connectivity +0.20
- Memantine: delta -0.10, theta -0.15, alpha +0.10, beta +0.05, connectivity +0.15

Band direction constraints operate on reconstructed versus baseline band-power blocks in the 2185-D vector (delta, theta, alpha, and beta channel blocks). For each band, the mean reconstructed change relative to baseline is compared to the prior sign: decreases are penalized when an increase is expected, and increases are penalized when a decrease is expected, scaled by the absolute prior weight. A separate connectivity constraint on coherence features penalizes mean coherence decreases under drug conditions. Baseline (no-drug) batches incur zero constraint terms. Exact feature-index slices are listed in Supplementary Material S1.

The total constrained objective is

$$L = L_{\text{MSE}} + \beta L_{\text{KL}} + \alpha_c L_{\text{band}} + \alpha_c L_{\text{conn}},$$

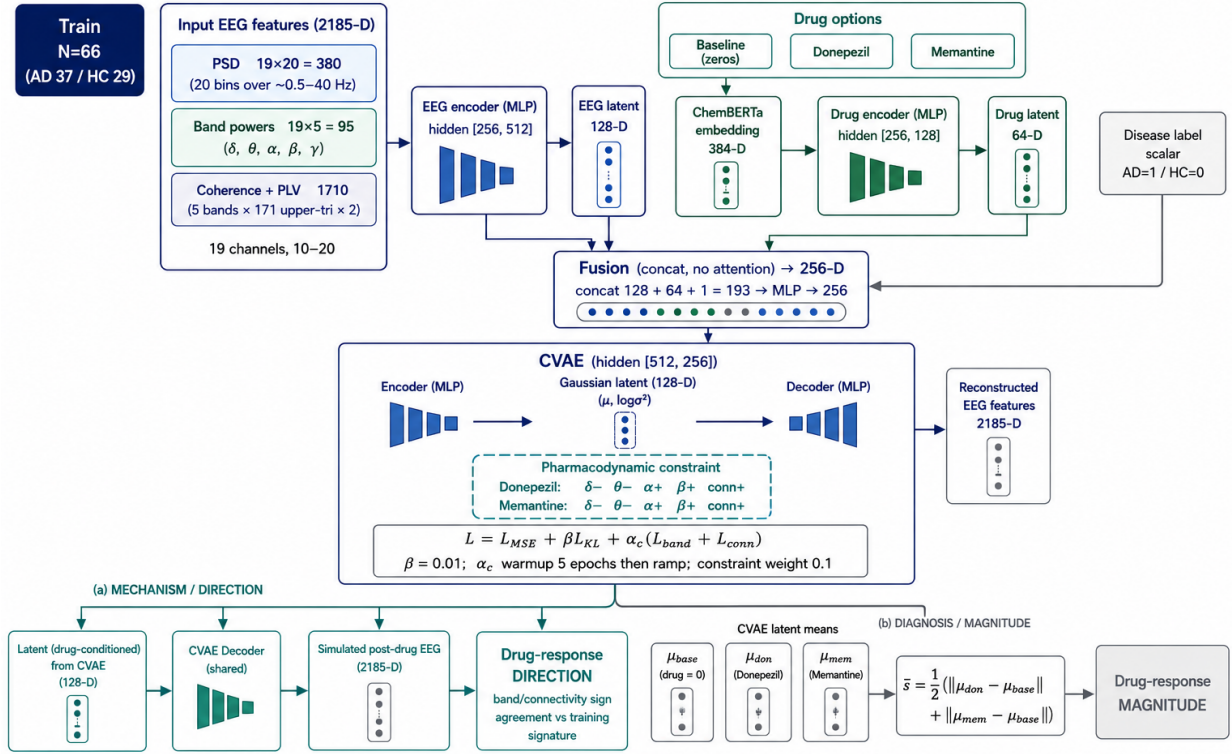


Figure 1: Architecture of the pharmacodynamically constrained conditional variational autoencoder (CVAE) digital twin. Resting-state EEG features (2185-D) and ChemBERTa drug embeddings are encoded and fused with a disease label, then mapped by a CVAE whose training includes pharmacodynamic band and connectivity constraints. Training cohort: $N = 66$ (AD 37 / HC 29).

with $\beta = 0.01$. Constraint weight $\alpha_c = 0$ for the first five warmup epochs, then ramps linearly over five epochs to the paper weight $\alpha_c = 0.1$. Constraints therefore bias reconstructed feature changes toward literature-signed pharmacology while reconstruction and KL terms remain active. This design prioritizes mechanistic consistency of simulated drug effects; any effect on diagnostic content of μ is treated as an empirical question for Results, not assumed here.

2.3. Training data and procedure

With architecture and constraints fixed, we describe the derivation cohort and optimization used for the frozen twin evaluated externally.

Training features come from resting-state EEG for 66 subjects (AD $n = 37$, HC $n = 29$). The derivation cohort is OpenNeuro accession ds004504 (Miltiadous et al. AD/HC resting EEG). Preprocessing used band-pass 0.5–40 Hz, notch 50 Hz, ICA on, 4 s epochs at 90% overlap, per-subject z-score, then spectral and connectivity features with time-averaged epochs before Welch (20 PSD bins at 500 Hz), as summarized in the domain-shift diagnosis notes.

The subject-level $N = 66$ training set is small relative to model capacity (2185-D inputs; CVAE latent dimension 128; multilayer encoders with widths up to 512). We therefore treat the twin as an externally validated mechanistic simulator rather than as a large-cohort estimator

Table 1: External EEG datasets used for escalating validation of the constrained twin. Sampling rates list native acquisition then the common 500 Hz analysis rate. Roles follow the Results narrative sequence.

Dataset	N	Groups	Labels	Sampling rate	Montage	Role
TUH	200	Abn 131 / Nor 69	Abnormal vs normal	Mixed EDF → 500 Hz	19ch 10–20	First external; proxy labels
OSF	92	AD 80 / HC 12	Folder AD vs HC	256 → 500 Hz	19ch 10–20	AD/HC; small HC
P-ADIC	145	AD 49 / HC 96	NIA-AA AD vs HC	200/500 → 500 Hz	19ch 10–20	Larger AD/HC; null check
CAUEEG	1122	Nor 436 / MCI 395 / Dem 291	Normal / MCI / Dementia	200 → 500 Hz	19ch AVG 10–20	Large-N; encoding

of population diagnosis, rely on subject-level five-fold training with a held-out validation split within each fold, and evaluate all primary claims on independent external cohorts rather than on training-set discrimination alone. Overfitting risk remains a limitation of the design and is revisited in the Discussion.

Constrained training used subject-level 5-fold cross-validation (five splits; random seed 42). Within each fold, the non-test subjects were split into train and validation (10% validation, stratified by label). Optimization used Adam with learning rate 1×10^{-4} , weight decay 1×10^{-5} , batch size 16, and 100 epochs on an NVIDIA GeForce RTX 5060 Ti (16 GB). The evaluation checkpoint for the main external battery is the fold-0 constrained model at epoch 99 with constraint weight 0.1 and five warmup epochs (alias `checkpoint_constrained.pt`). External twin simulations in this paper use that checkpoint with five Monte Carlo draws per condition unless a fold-sensitivity analysis is stated.

Checkpoint selection statement. The analysis used the documented fold-0 checkpoint selection rule (epoch 99; $\alpha_c = 0.1$; alias `checkpoint_constrained.pt`), matched to the unconstrained control and treated as the locked paper checkpoint. The available records document that selection rule but do not independently establish the historical chronology of external-metric inspection. After the external battery was defined, cohort results were not used to retune α_c , fold index, or epoch for the manuscript tables. The α_c sweep in Section 3.5 is an exploratory sensitivity analysis on fold-0 twins and does not redefine the locked paper weight.

2.4. External validation datasets

Training established the twin. External datasets were introduced in sequence so each cohort answered a specific doubt left by the previous magnitude or label setting. Cohort size, labels, sampling rate, montage, and study role are summarized in Table 1.

Temple University Hospital (TUH) clinical EEG. We used 200 successfully processed EDFs. Labels for magnitude tests were header/path-derived abnormal versus normal ($n = 131$ abnormal, $n = 69$ normal), not AD-specific diagnoses. Channels were mapped to 19 legacy 10-20 names, recordings cropped to 320 s, resampled to 500 Hz, notch 60 Hz (US mains), ICA off for tractability, with epoch-count spectral calibration to 864 epochs. TUH was selected first to test whether latent drug-response magnitude discriminated on independent hospital EEG after leaving the training resting-state distribution.

Open Science Framework (OSF) Eyes_closed AD/HC. After the TUH magnitude null under proxy labels, we used folder-labeled AD ($n = 80$) and Healthy/HC ($n = 12$) features. Native rate was 256 Hz with 19-channel 10-20 layout, resampled to 500 Hz; stored processed shape implied one 4 s epoch per subject before calibration to the same 864-epoch spectral reference. OSF was selected to ask whether true AD versus HC labels (still research partitions, not NIA-AA gold standard in local metadata) would restore magnitude discrimination that abnormal/normal proxies lacked.

P-ADIC Dryad AD/HC. After OSF’s positive but HC-underpowered result, we used Dryad release recordings resolved to AD $n = 49$ and HC $n = 96$ at recording level, DOI 10.5061/dryad.8gtht76pw [26]. Arrays were remapped from an assumed Nihon-order montage into training channel order, cropped to 320 s, resampled to 500 Hz when needed, notch 50 Hz, and calibrated to 864 epochs. P-ADIC was selected to test whether a larger NIA-AA-oriented AD/HC resource confirmed OSF or exposed fragility of the HC $n = 12$ result.

CAUEEG (Chung-Ang University Hospital EEG). After P-ADIC’s magnitude null, we used the dementia-no-overlap benchmark pool (train+validation+test) with Normal $n = 436$, MCI $n = 395$, Dementia $n = 291$ (1122/1122 features OK). EDFs are 19-channel average-referenced 10-20 at 200 Hz plus non-EEG channels dropped by mapping; processing used notch 60 Hz (South Korea), crop 320 s, resample 500 Hz, and spectral calibration to 864 epochs. Band-power scale versus training was ratio 1.104. CAUEEG was selected to test magnitude discrimination on large clinical Normal/MCI/Dementia strata with calibrated spectral scale and good PCA overlap, and to host the encoding probes.

Direction agreement for mechanism tests used TUH, OSF, and P-ADIC cohort-mean drug-minus-baseline feature effects against the training-cohort mean effects (CAUEEG was not required for the locked 10/10 direction claim).

2.5. Evaluation endpoints

Datasets alone do not define success. We therefore fix three endpoint families used uniformly in Results. **No paired pre/post-dose EEG** is available in the external cohorts; the direction endpoint measures agreement with the **training constrained signature**, not empirical pharmacodynamic change after drug administration.

Primary hierarchy. P1 (primary): effect-vector Pearson correlation r between each external cohort’s mean Donepezil/Memantine drug-minus-baseline feature vector and the training constrained signature (with cosine similarity as a secondary similarity check). Block-level signed agreement (10/10 maximum) is reported as an interpretable but coarse secondary summary, because a prior-only constant effect can also reach perfect signs. P2: CAUEEG Dementia versus Normal magnitude on $\bar{s}$. P3: CAUEEG latent probe ROC-AUC on μ_{base} versus spectral and raw-feature references. Remaining contrasts (TUH proxy labels, OSF magnitude with HC $n = 12$, P-ADIC magnitude, nested heads, α_c sweep) are escalation or exploratory follow-ups.

Analysis roles. External signature concordance (P1 on the locked fold-0 checkpoint) is the primary endpoint. Five-fold scoring, prior-only, matched unconstrained, and equal-weight ensemble analyses are robustness or control analyses. Diagnostic magnitude and encoding probes are secondary. The α_c sweep, domain-shift classifier, feature-family forensics, and related checks

are exploratory and are not treated as a single confirmatory multiplicity family. No multiplicity adjustment was applied across this hierarchy.

Drug-response direction agreement. For each external cohort with saved per-subject simulations, we form the mean Donepezil-minus-baseline and Memantine-minus-baseline vectors in 2185-D feature space and compare each to the corresponding mean effect from the training-cohort constrained simulations. Agreement is counted on five blocks: delta, theta, alpha, beta, and connectivity. A block agrees if $\text{sign}(\text{mean}(a_{\text{block}})) = \text{sign}(\text{mean}(b_{\text{block}}))$, or both absolute means are below 10^{-8} . Donepezil and Memantine each contribute five checks, so perfect agreement is 10/10. The primary continuous score is the mean of Donepezil and Memantine Pearson correlations between those full mean effect vectors. Uncertainty for r uses subject-level bootstrap percentile intervals (2000 resamples of subjects with replacement; seed 42), not Fisher z intervals that treat the 2185 features as independent observations. Cosine similarity of the same mean effect vectors is reported alongside r .

Drug-response magnitude. For each subject, after encoding, we compute

$$s_{\text{don}} = \|\mu_{\text{don}} - \mu_{\text{base}}\|_2, \quad s_{\text{mem}} = \|\mu_{\text{mem}} - \mu_{\text{base}}\|_2, \quad \bar{s} = \frac{1}{2}(s_{\text{don}} + s_{\text{mem}}).$$

Group differences on $\bar{s}$ are the diagnostic magnitude tests (abnormal vs normal; AD vs HC; Dementia vs Normal).

Encoding probes (CAUEEG only). Baseline latents were extracted with drug vector all zeros and disease label fixed to 0 for every subject to avoid leaking diagnosis into fusion. Primary analysis used Dementia versus Normal only ($n = 727$; MCI excluded). A logistic regression probe used 5-fold stratified cross-validation (shuffle on; random state 42), with features standardized using training-fold statistics only and class-balanced logistic regression (maximum 2000 iterations). A 200-label-permutation null repeated the same 5-fold probe. Reference probes used the same folds on (i) scalar theta/alpha ratio from band powers and (ii) the full 2185-D feature vector. Stronger heads used nested CV: outer 5-fold stratified, inner 3-fold grid search on ROC-AUC for tuned L2 logistic regression, MLP, and histogram gradient boosting, plus a μ concatenated with log-variance control for the best head. The best head’s out-of-fold ROC-AUC was tested with 200 permutations of the same nested protocol.

Ablations and sensitivity. A matched unconstrained fold-0 checkpoint (same selection rule as the constrained fold-0 twin), a prior-only signed effect baseline, and an exploratory constraint-weight (α_c) fold-0 sweep test whether direction preservation and latent diagnostic AUC move with constraint strength. Separately, all five constrained fold checkpoints were scored on TUH, OSF, and P-ADIC direction against the same training signature to quantify fold-to-fold variability of the primary r endpoint. An equal-weight average of the five fold effect vectors (subject-level ensemble) is reported only as a robustness check against that fold variance and against the prior-only scalar; it is not used for checkpoint selection.

2.6. Statistics

Endpoints in Section 2.5 map onto the following tests.

Two-group magnitude comparisons used two-sided Mann-Whitney U tests and Cohen’s d with sample variance (degrees of freedom $n - 1$). Approximate 95% confidence intervals on Cohen’s

d in Fig. 3 use the Hedges-Olkin standard-error formula from the reported d and group sizes; they are not bootstrap or exact intervals stored in the validation artifacts. CAUEEG also reported a three-group Kruskal-Wallis test on Normal, MCI, and Dementia magnitude scores. Encoding probes reported ROC-AUC, balanced accuracy, and F1 per fold and as means with sample standard deviation across folds, plus out-of-fold aggregates. Where both are shown, mean-fold AUC is the average of fold-wise AUCs and OOF AUC is the pooled out-of-fold score; for the theta/alpha reference these are 0.675 and 0.673, respectively. Effect-vector r uncertainty uses subject-level bootstrap percentile intervals as defined in Section 2.5. Permutation p-values used

$$p = \frac{1 + \#\{\text{null AUC} \geq \text{true AUC}\}}{N_{\text{perm}} + 1}$$

with $N_{\text{perm}} = 200$. Nested head selection optimized inner-fold ROC-AUC only; outer test folds were never used for hyperparameter choice.

No multiplicity correction (Bonferroni, FDR, or similar) was applied across the escalating external contrasts or across probe metrics. Each primary contrast was treated as a pre-specified question in the escalation sequence (proxy hospital EEG; AD/HC small HC; larger AD/HC; large clinical dementia spectrum; then encoding localization), and secondary contrasts (for example MCI pairs, disease-label ablation) are reported descriptively.

Hardware for feature extraction, constrained training, and the 1122-subject CAUEEG twin pass used the project Python 3.10 environment on an NVIDIA GeForce RTX 5060 Ti (16 GB).

These definitions and datasets are used without change in the Results that follow, beginning with constraint-consistent direction agreement on the three external cohorts that support the 10/10 signature claim.

3. Results

3.1. Matched unconstrained and prior-only baselines

Section 2.5 defined constraint-consistent direction agreement against the training constrained signature and reserved matched unconstrained and prior-only checks for interpreting that endpoint. We report those controls first, then the constrained external replications.

Matched unconstrained twin. The unconstrained checkpoint uses the same fold-0 selection rule as the constrained external twin (Methods; Supplementary Material S1). On the same direction endpoint and constrained training signature, unconstrained agreement was 3/10 on TUH (effect-vector $r = -0.309$), 4/10 on OSF ($r = -0.331$), and 4/10 on P-ADIC ($r = -0.327$). Constrained agreement on the same three cohorts is 10/10 (Section 3.2). Magnitude contrasts for the unconstrained twin remained null or non-diagnostic on TUH abnormal versus normal ($p = 0.124$, $d = -0.04$), OSF AD versus HC ($p = 0.124$, $d = -0.16$), P-ADIC AD versus HC ($p = 0.791$, $d = 0.03$), and CAUEEG Dementia versus Normal ($p = 0.088$, $d = -0.07$). On CAUEEG Dementia versus Normal ($n = 727$), the unconstrained μ_{base} logistic probe reached mean ROC-AUC = 0.539, below the constrained latent probe (0.579) and below the theta/alpha spectral reference on the same subjects (mean-fold ROC-AUC = 0.675). Table 2 summarizes constrained versus unconstrained direction on TUH, OSF, and P-ADIC.

Table 2: Matched unconstrained versus constrained direction agreement against the constrained training signature. Prior-only signed effects also reach 10/10 on signs with lower mean effect-vector correlation ($r=0.636$).

Dataset	n	Constr. agree	Constr. r	Unconstr. agree	Unconstr. r
TUH	200	10/10	0.908	3/10	-0.309
OSF	92	10/10	0.851	4/10	-0.331
P-ADIC	145	10/10	0.918	4/10	-0.327

Prior-only baseline. Applying literature signed band and connectivity deltas as a constant 2185-D effect vector, without a CVAE, yielded 10/10 direction agreement against the constrained training signature, with mean effect-vector correlation $r = 0.636$ (Donepezil $r = 0.572$; Memantine $r = 0.700$). The prior-only baseline therefore shows that directional agreement is partly recoverable from the literature constraint itself. Locked fold-0 constrained correlations ($r = 0.851$ to 0.918 ; Section 3.2) are substantially higher in continuous similarity, but those increments should not be read as evidence that the model independently discovered the drug direction. Rather, at the selected checkpoint the learned constrained response structure more closely reproduces the training signature than the constant prior alone. The five-fold ensemble does not clearly exceed the prior (Section 3.2), so the prior remains a binding reference for fold-averaged continuous agreement.

These controls establish two facts used below. First, matched unconstrained training does not preserve the constrained signature under external domain shift. Second, prior-only signs explain much of the binary agreement while leaving room for fold-0 continuous similarity above the prior, without licensing ensemble-level superiority over the prior. Whether raising α_c locks direction and how latent AUC moves is deferred to Section 3.5.

3.2. Constrained twin: drug-response direction agreement

Having established that unconstrained training fails the direction endpoint, we ask whether the constrained twin reproduces the training cohort’s signed band and connectivity effects on independent EEG. The **primary** score is effect-vector correlation r (mean of Donepezil and Memantine Pearson correlations). Block-level signed agreement (maximum 10/10) is a secondary, coarse summary. Because the external cohorts do not contain paired pre/post-drug EEG measurements, this analysis evaluates concordance with the learned training signature rather than empirical post-dose pharmacodynamic change.

Across three external cohorts, **locked fold-0** constrained effect-vector correlations were $r = 0.908$ (TUH), 0.851 (OSF), and 0.918 (P-ADIC), with secondary sign agreement 10/10 in every case. These high continuous correlations are specific to the locked fold-0 checkpoint; five-fold sensitivity below shows substantially lower mean continuous similarity. Per-dataset values are summarized in Table 3 and illustrated in Fig. 2.

Subject-bootstrap 95% percentile intervals for r (2000 subject resamples) are TUH 0.874 to 0.900 (boot mean 0.888; $n = 200$), OSF 0.790 to 0.837 (boot mean 0.815; $n = 92$), and P-ADIC 0.871 to 0.904 (boot mean 0.889; $n = 145$). Mean cosine similarities of the same effect vectors

Table 3: Locked fold-0 band-level drug-response direction agreement between each external cohort and the training-cohort constrained signature. Agreement is scored on five feature blocks $\times$ two drugs (Donepezil, Memantine; maximum 10/10). Correlation r is the Pearson correlation of mean drug-minus-baseline effect vectors (effect_magnitude_correlation). These are the paper checkpoint values, not five-fold re-simulation fold-0 scores. Replication status follows the Results escalating-evidence sequence. CAUEEG is not included in the locked 10/10 direction claim.

Dataset	N	Agree	r	Replication status
TUH	200	10/10	0.908	1st confirmation
OSF	92	10/10	0.851	2nd confirmation (independent)
P-ADIC	145	10/10	0.918	3rd confirmation (acq. change)

are 0.936, 0.896, and 0.942. Fisher- z intervals that treat feature length 2185 as an independent-observation n are not used for inference.

TUH (first confirmation). On Temple University Hospital clinical EEG ($n = 200$), Donepezil and Memantine each agreed on all five blocks (5/5 and 5/5; total 10/10), with overall effect-vector correlation $r = 0.908$. A falsifying outcome would have been any band or connectivity sign flip relative to the training signature, or a near-zero correlation. Neither occurred. Relative to the matched unconstrained twin (3/10 on the same endpoint), this is the first external confirmation that constraint-consistent response-direction structure can survive hospital EEG outside the training resting-state distribution at the locked fold-0 checkpoint.

OSF (second confirmation). On the OSF Eyes_closed AD/HC set ($n = 92$), direction agreement was again 10/10 (Donepezil 5/5, Memantine 5/5), with $r = 0.851$. This cohort changes acquisition and label structure relative to TUH. Unconstrained agreement on the same OSF features was 4/10. Agreement for the constrained twin remained perfect.

P-ADIC (third confirmation). On P-ADIC Dryad AD/HC recordings ($n = 145$), direction agreement was 10/10 (Donepezil 5/5, Memantine 5/5), with $r = 0.918$. Unconstrained agreement was 4/10. Constrained agreement remained perfect after montage remapping and spectral calibration.

Fig. 2 shows the three constrained effect-vector correlations with the corresponding 10/10 annotations for the locked fold-0 paper checkpoint. Table 3 lists n , agreement fractions, r , and replication status for each cohort. Table 2 places the matched unconstrained failures beside those constrained successes.

Five-fold sensitivity. Scoring all five constrained fold checkpoints on the same TUH / OSF / P-ADIC protocol yields mean $r = 0.385 \pm 0.308$ (TUH), 0.371 ± 0.294 (OSF), and 0.372 ± 0.310 (P-ADIC). Sign agreement was 10/10 on three folds, 9/10 on fold 4, and 8/10 on fold 2. Continuous effect-vector similarity is therefore fold-sensitive: the locked fold-0 checkpoint recovers high r (TUH 0.897, OSF 0.850, P-ADIC 0.900 in the five-fold re-simulation), whereas several other folds preserve signs more than quantitative structure. Main-text mechanistic claims use the pre-locked fold-0 alias; the five-fold spread is reported as a sensitivity check, not as an ensemble claim.

Equal-weight fold ensemble (robustness check). Averaging Donepezil and Memantine

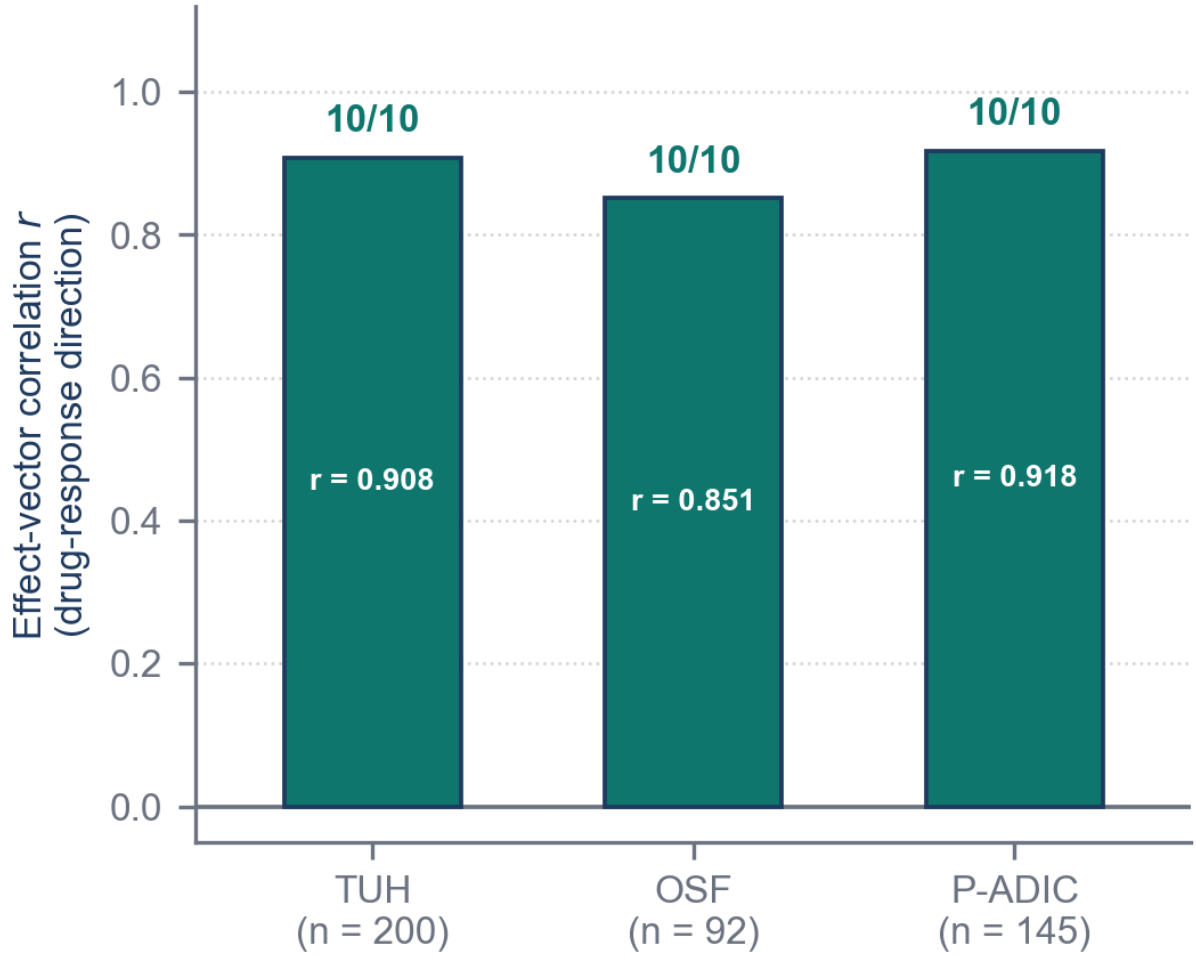


Figure 2: External constrained drug-response direction agreement for the locked fold-0 checkpoint. Bars show effect-vector correlation r for TUH ($r=0.908$, $n=200$), OSF ($r=0.851$, $n=92$), and P-ADIC ($r=0.918$, $n=145$). Secondary band-level signed agreement was 10/10 on each cohort. Subject-bootstrap 95% percentile intervals for r are reported in the text.

subject-level effect vectors equally across folds 0 to 4, then scoring cohort-mean effects against the fixed training signature, yields $r = 0.642$ on TUH (subject-bootstrap 95% percentile interval 0.618 to 0.645; $B = 2000$, seed 42), $r = 0.626$ on OSF (0.584 to 0.624), and $r = 0.627$ on P-ADIC (0.595 to 0.629). The ensemble provides a fold-averaged robustness assessment, but the available evidence does not establish incremental agreement beyond the prior-only baseline (mean $r = 0.636$; paired Δr under shared subject resampling is not supported by stored prior-only artifacts). Ensembling does not restore fold-0 continuous similarity and does not license claiming superiority over the prior.

These results establish that the pharmacodynamically constrained twin can preserve constraint-consistent drug-response *structure* across three independent external EEG resources when the locked fold-0 checkpoint is used, with secondary perfect signs and high r . Continuous fidelity is

Table 4: Twin latent drug-response magnitude discrimination across escalating external cohorts. Status strings are taken from the validation JSON (FAIL, PASS_WITH_CAVEATS). The OSF primary row uses `honest_verdict.status`; OSF disease-label sensitivity rows have no status field (shown as –). The Kruskal–Wallis row reports H instead of Cohen’s d and Kruskal p in the p column.

Dataset	Contrast	N (A / B)	p	d	Status
TUH					
TUH	Abnormal vs normal	131 / 69	0.272	-0.13	FAIL
OSF					
OSF	AD vs HC	80 / 12	$p < 0.001$	0.44	PASS*
OSF	AD vs HC (disease=0)	80 / 12	$p < 0.001$	0.64	–
OSF	AD vs HC (disease=1)	80 / 12	$p < 0.001$	0.34	–
P-ADIC					
P-ADIC	AD vs HC	49 / 96	0.968	-0.07	FAIL
P-ADIC	AD vs HC (disease=0)	49 / 96	0.527	-0.16	FAIL
P-ADIC	AD vs HC (disease=1)	49 / 96	0.702	-0.14	FAIL
CAUEEG					
CAUEEG	Dementia vs Normal	291 / 436	0.516	-0.00	FAIL
CAUEEG	Dementia vs MCI	291 / 395	0.877	0.07	FAIL
CAUEEG	MCI vs Normal	395 / 436	0.363	-0.07	FAIL
CAUEEG	AD-tagged Dem. vs Normal	214 / 436	0.624	-0.01	FAIL
CAUEEG	Kruskal–Wallis (3 groups)	436 / 395 / 291	0.637	$H=0.90$	FAIL
CAUEEG	Dem. vs Normal (disease=0)	291 / 436	0.411	-0.04	FAIL
CAUEEG	Dem. vs Normal (disease=1)	291 / 436	0.776	0.00	FAIL

*PASS_WITH_CAVEATS (HC $n=12$; exploratory).

highly fold-dependent, and fold-averaged continuous agreement does not clearly exceed the prior. Whether this signature concordance translates into diagnostic utility of drug-response *magnitude* is a separate question, addressed next. Section 3.2 showed that constraint-consistent drug-response *direction* agrees with the training signature at 10/10 on three cohorts, while Section 3.1 showed that a matched unconstrained twin fails that endpoint. We now ask whether drug-response *magnitude* $\bar{s}$ carries external diagnostic signal under the Section 2.5 definition. The investigation is sequential: each cohort answers a doubt left by the previous null or caveated result. Primary contrasts and secondary checks are summarized in Table 4. Effect sizes with approximate 95% confidence intervals are illustrated in Fig. 3.

TUH: first external magnitude test (proxy labels). On TUH clinical EEG we compared abnormal ($n = 131$) versus normal ($n = 69$) on $\bar{s}$. The contrast was null (Mann-Whitney $p = 0.272$, Cohen’s $d = -0.13$; status FAIL). Labels are header/path-derived abnormal versus normal, not AD-specific diagnoses, and the processed cohort carried no AD-drug annotations usable as response outcomes. A credible confound for this null is therefore weak proxy labeling rather than a definitive failure of the twin under true AD/HC strata. The open question is whether AD versus HC labels restore magnitude discrimination.

OSF: AD/HC labels with a thin control arm. On OSF Eyes_closed AD ($n = 80$) versus HC ($n = 12$), $\bar{s}$ was larger in AD ($p = 0.000278$, $d = 0.44$; discrimination status PASS; exploratory

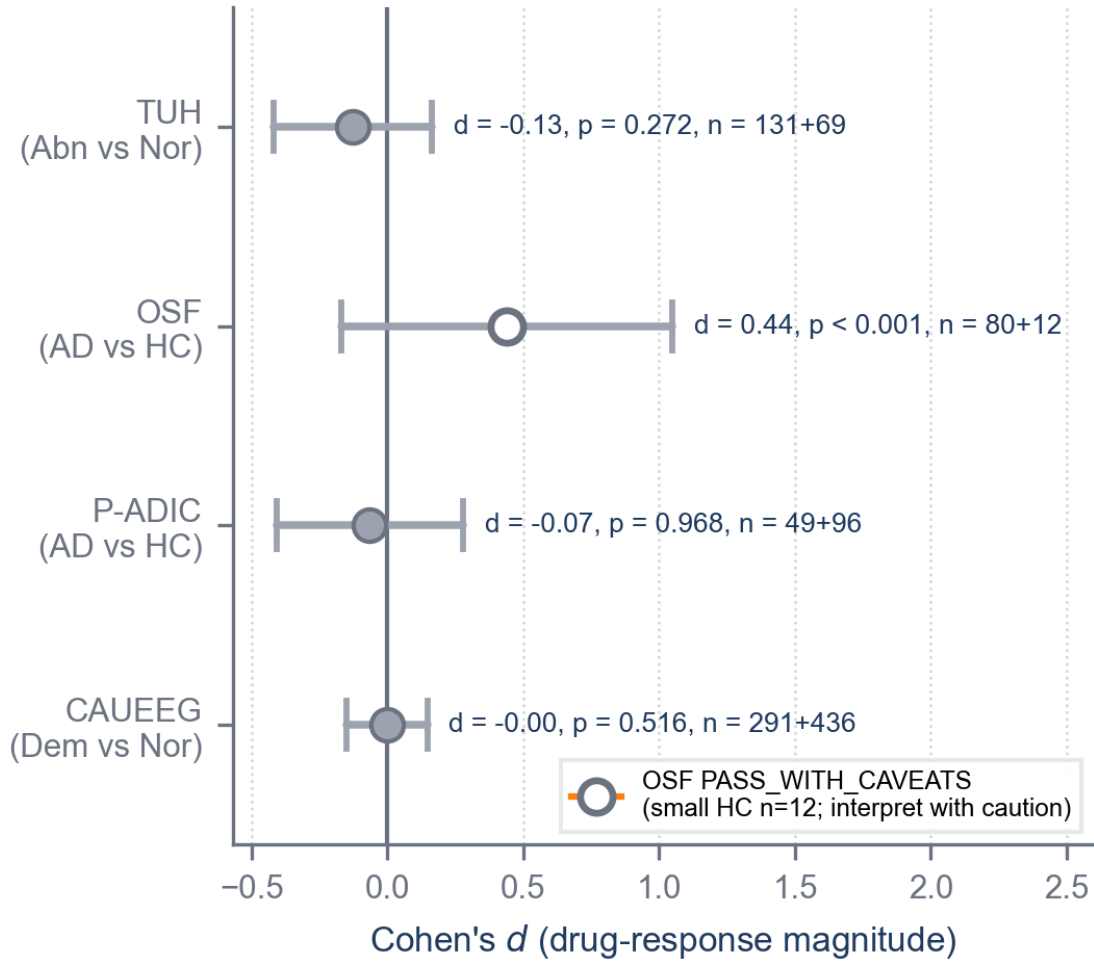


Figure 3: Forest plot of Cohen's d for primary twin magnitude contrasts on TUH, OSF, P-ADIC, and CAUEEG. Approximate 95% confidence intervals use the Hedges–Olkin standard-error formula from reported d and group sizes (not bootstrap intervals stored in validation artifacts). Exact n , p , and d values are listed in Table 4.

PASS_WITH_CAVEATS in the validation report). Separation persisted when disease conditioning was fixed for all subjects (disease = 0: $p < 0.001$, $d = 0.64$; disease = 1: $p < 0.001$, $d = 0.34$), so the signal is not an artifact of feeding diagnosis into fusion alone. The result is not a clean external PASS: HC $n = 12$ limits power, and the absolute mean gap is small. The open question is whether a larger, clinically structured AD/HC resource confirms OSF or exposes fragility of the HC arm.

P-ADIC: larger AD/HC, null magnitude. On P-ADIC recordings (AD $n = 49$, HC $n = 96$), $\bar{s}$ did not separate groups ($p = 0.968$, $d = -0.07$; status FAIL). Disease-label ablation remained null (disease = 0: $p = 0.527$, $d = -0.16$; disease = 1: $p = 0.702$, $d = -0.14$; both FAIL). Age imbalance (AD mean age about 70 years versus HC about 52 years in the release metadata) is a listed confound, but age would typically inflate rather than erase AD/HC differences if it drove $\bar{s}$; here differences are absent. The open question is whether large- N hospital Normal/MCI/Dementia strata with calibrated spectral scale still show a magnitude null.

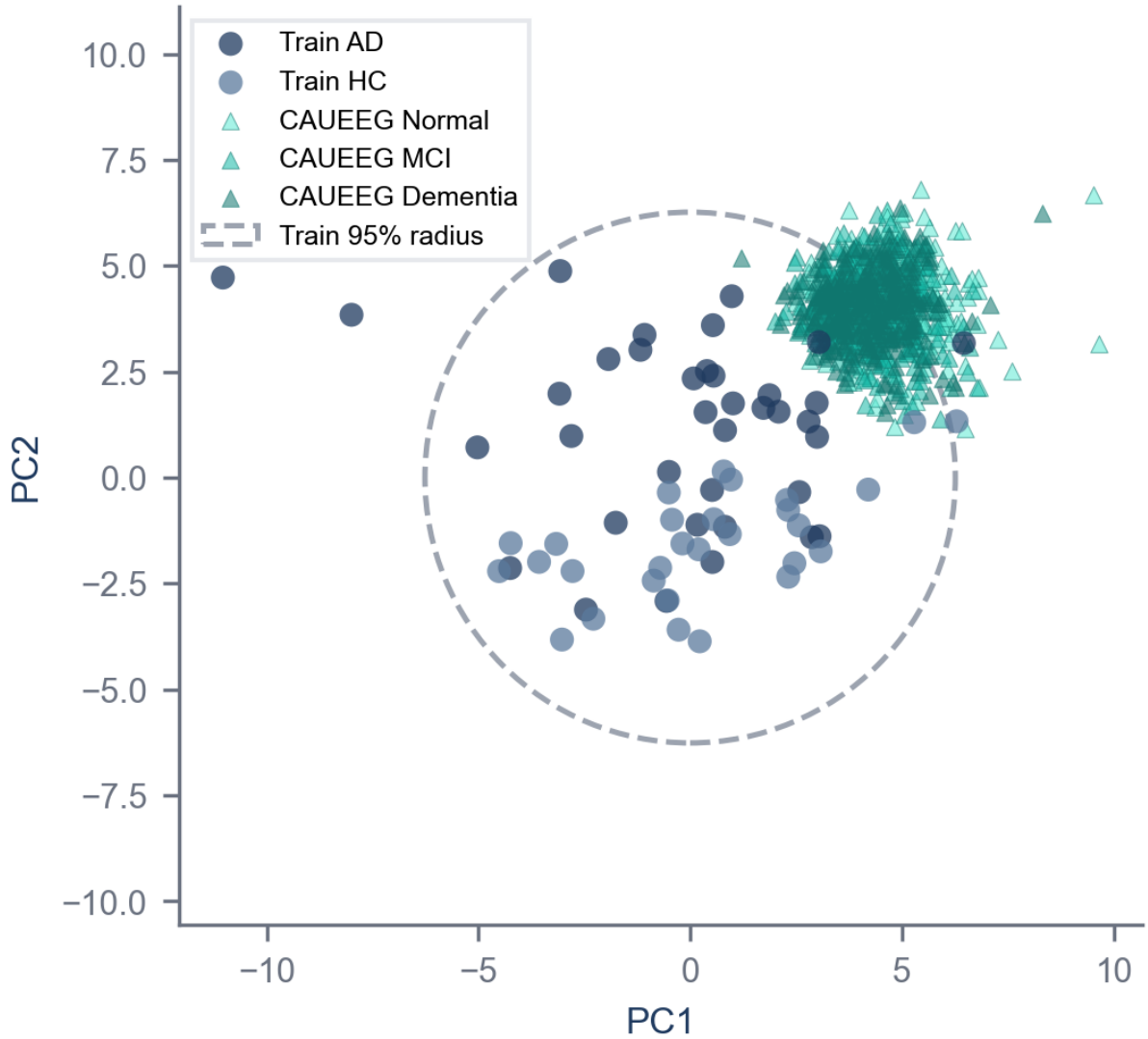


Figure 4: PCA of 2185-D features: training cohort versus CAUEEG. Dashed circle marks the training 95% radius; overlap fraction within that radius is 0.611.

CAUEEG: large clinical strata and a domain check. CAUEEG was selected next because it supplies large clinical Normal ($n = 436$), MCI ($n = 395$), and Dementia ($n = 291$) pools (1122 features OK), with band-power scale near training (ratio 1.104) and good PCA overlap with the training cohort (overlap fraction within the training 95% radius = 0.611; Fig. 4).

Primary Dementia versus Normal on $\bar{s}$ was null ($n = 291 / 436$; $p = 0.516$, $d = -0.00$; status FAIL). Secondary twin contrasts were likewise null: Dementia versus MCI ($p = 0.877$, $d = 0.07$), MCI versus Normal ($p = 0.363$, $d = -0.07$), AD-tagged Dementia versus Normal ($n = 214 / 436$; $p = 0.624$, $d = -0.01$), and Kruskal-Wallis across three groups ($H = 0.90$, $p = 0.637$). Disease-label ablation on Dementia versus Normal remained FAIL (disease = 0: $p = 0.411$, $d = -0.04$; disease

= 1: $p = 0.776$, $d = 0.00$). Critically, the same CAUEEG features do carry clinical signal outside the twin endpoint: theta/alpha ratio Dementia versus Normal was strongly separated ($p < 0.001$, $d = 0.47$). The magnitude null is therefore not explained by absent clinical EEG structure in the inputs, nor by a failed scale/PCA check.

Fig. 3 places the four primary magnitude contrasts on a common Cohen’s d axis. Approximate 95% intervals use the Hedges-Olkin standard-error formula from reported d and group sizes (Methods). Table 4 records every listed contrast with exact n , p , d , and status strings from the validation runs.

Section 3.2 established constraint-consistent direction agreement. Section 3.3 shows that the twin’s magnitude endpoint does not diagnose reliably across an escalating sequence of external tests, despite PASS_WITH_CAVEATS on a small-HC OSF partition and despite clear spectral slowing in CAUEEG features. Clinical signal is present in raw features, but the diagnostic magnitude endpoint does not use it. The next section localizes residual clinical information at encoding versus readout. Section 3.3 left an explicit puzzle: CAUEEG features separate Dementia from Normal on theta/alpha slowing, yet the twin’s drug-response magnitude endpoint does not. We therefore ask whether clinical signal survives CVAE encoding at all, and if so, whether a more flexible readout can recover the feature-level references.

Baseline latents μ_{base} were extracted with drug vector zero and disease label fixed to 0 for every subject (Dementia versus Normal only; $n = 727$), so the probe cannot read diagnosis from the fusion disease bit. A linear logistic probe on μ_{base} reached mean cross-validated ROC-AUC = 0.579 (permutation $p = 0.010$ under 200 label shuffles of the same 5-fold protocol). The probe therefore beats the permutation null: measurable diagnostic information survives encoding. It does not match the input references. On the same folds, scalar theta/alpha ratio reached mean-fold ROC-AUC = 0.675 (pooled OOF 0.673), and a class-balanced logistic probe on the full 2185-D feature vector reached mean-fold ROC-AUC = 0.699 (OOF 0.699). Theta/alpha is therefore an interpretable spectral **reference**, not a proven feature ceiling: the packed 2185-D vector still exceeds it. The latent probe sits about 0.096 below theta/alpha and about 0.120 below the 2185-D reference.

If the gap were only a weak linear readout, nested stronger heads on frozen μ_{base} should close it. They did not. Out-of-fold ROC-AUC values were 0.593 for tuned L2 logistic regression (best head; permutation $p = 0.005$ under 200 nested-CV shuffles), 0.543 for MLP, 0.556 for HistGradientBoosting, and 0.587 for a μ +logvar control. Under the tested readout functions, performance appeared to saturate near the linear probe (stored nested-head artifact tag: readout saturation). The best head remains near the original linear probe and well below both feature references. The pattern is consistent with an encoding-level information bottleneck: diagnostic information is substantially attenuated by the learned latent representation under the tested readouts, rather than recovered by readout complexity alone.

Fig. 5 places the spectral reference (mean-fold AUC = 0.675), the original latent probe, and all nested heads on a common ROC-AUC axis against the latent-probe permutation null band (5th to 95th percentiles 0.449 to 0.559) and chance (AUC = 0.5).

Table 5 reports the corresponding out-of-fold (OOF) AUCs, balanced accuracies, permutation p -values where computed, and signed gaps versus the theta/alpha OOF reference (0.673). The 2185-D OOF AUC of 0.699 is cited in the text as the fuller feature reference.

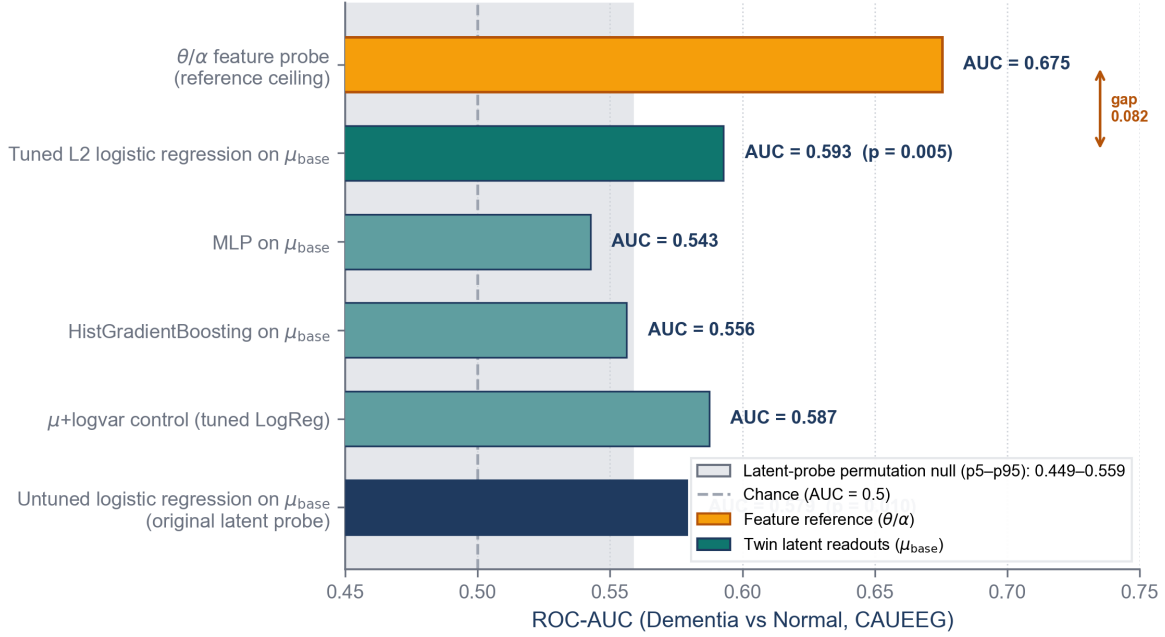


Figure 5: Encoding analysis on CAUEEG Dementia versus Normal ($n=727$). Amber: theta/alpha feature probe (mean-fold ROC-AUC = 0.675; pooled OOF = 0.673). Teal/navy: frozen-twin μ_{base} readouts. Gray band: latent-probe permutation null 5th–95th percentiles (0.449–0.559). Best nested head OOF AUC = 0.593 remains below the spectral reference; a separate 2185-D logistic reference reaches mean-fold AUC = 0.699.

Table 5: CAUEEG Dementia vs Normal encoding analysis ($n = 727$). Out-of-fold (OOF) ROC-AUC and balanced accuracy are reported. Permutation p -values are included where computed (original latent probe: 200 label shuffles of 5-fold CV; best head: 200 shuffles of nested CV). The final column is OOF AUC minus the θ/α pooled OOF AUC (0.673; mean-fold θ/α AUC = 0.675 is used in the prose and Fig. 5). Complements Fig. 5.

Model	Input	AUC	Bal. acc.	Perm. p	vs. θ/α
θ/α probe (ref.)	θ/α	0.673	0.625	–	+0.000
Untuned logistic	μ_{base}	0.576	0.568	0.010	-0.097
Tuned L2 logistic	μ_{base}	0.593	0.573	0.005	-0.080
MLP	μ_{base}	0.543	0.543	–	-0.130
HistGradientBoosting	μ_{base}	0.556	0.548	–	-0.117
Tuned LogReg (μ +logvar)	μ +logvar	0.587	0.554	–	-0.085

Section 3.1 already showed that a matched unconstrained encoder does not raise CAUEEG latent AUC above the constrained probe (unconstrained mean AUC = 0.539 versus constrained 0.579). Encoding loss relative to the spectral reference is therefore not explained as a simple unconstrained-versus-constrained compression on this comparison. The next section asks whether sweeping constraint weight α_c moves direction fidelity and latent AUC together. Sections 3.1 to 3.4 established that constraints are required for external signature preservation, that magnitude fails diagnostically, and that μ_{base} retains partial clinical signal below feature references. We now vary the constraint weight α_c on matched fold-0 twins (same selection rule as the paper checkpoint) and score TUH and P-ADIC direction against the constrained training signature plus CAUEEG

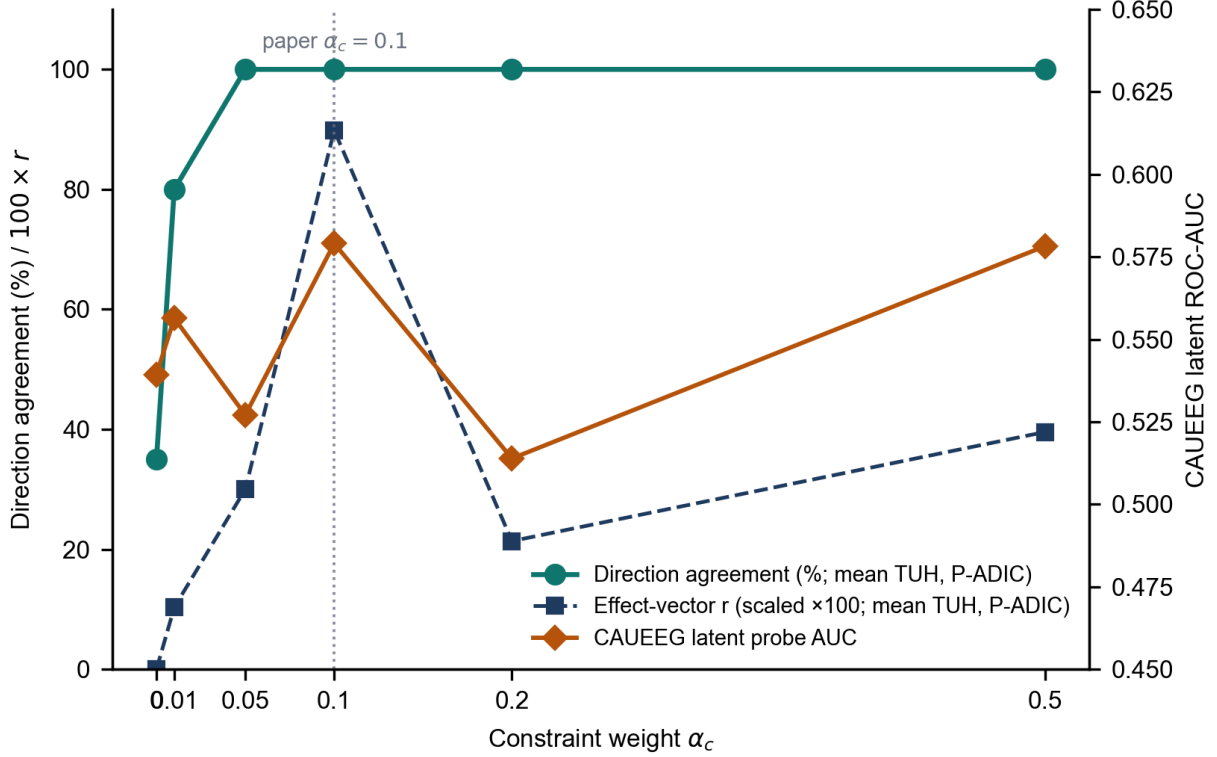


Figure 6: Constraint-weight sweep on matched fold-0 twins. Left axis: mean TUH/P-ADIC direction agreement (%) (teal) and 100 $\times$ mean effect-vector r (navy). Right axis: CAUEEG Dementia versus Normal latent-probe ROC-AUC (amber). Direction signs lock by $\alpha_c=0.05$; effect-vector r peaks at the paper weight $\alpha_c=0.1$; latent AUC is non-monotone. Sweep uses fold-0 twins and TUH/P-ADIC direction scores (not full 5-fold CV; OSF omitted from this sweep).

Dementia versus Normal latent-probe AUC. This is an **exploratory** constraint-strength sensitivity analysis, not a nested CV of α_c . Results are illustrated in Fig. 6.

At $\alpha_c = 0$ (seeded unconstrained fold-0), direction agreement was 3/10 on TUH (effect-vector $r = -0.350$) and 4/10 on P-ADIC ($r = -0.364$), with CAUEEG latent mean AUC = 0.539. At $\alpha_c = 0.01$, agreement rose to 8/10 on both TUH and P-ADIC, but effect-vector correlations remained low (TUH $r = 0.096$). At $\alpha_c = 0.05$, agreement reached 10/10 on both cohorts (TUH $r = 0.299$; latent AUC = 0.527). At the paper weight $\alpha_c = 0.1$, agreement remained 10/10 and effect-vector correlation peaked (TUH $r = 0.897$; P-ADIC direction 10/10; latent AUC = 0.579). At $\alpha_c = 0.2$ and 0.5, direction signs stayed at 10/10 while effect-vector r fell (TUH $r = 0.220$ and 0.412) and latent AUC was 0.514 and 0.578, respectively.

Fig. 6 therefore shows a clear α_c dependence for signed direction agreement and a non-monotone effect-vector correlation that peaks at the paper weight. Latent diagnostic AUC does not fall as direction rises: the curve is non-monotone and does not support a simple direction-versus-diagnosis trade-off across α_c alone. Binary sign agreement also saturates earlier than continuous fidelity (10/10 by $\alpha_c = 0.05$ while TUH r continues to rise through 0.1), so 10/10 alone is an insufficient summary of response structure. The constrained twin’s external behavior is better

summarized as constraint-dependent signature preservation with a preferred operating weight near $\alpha_c = 0.1$ for effect-vector correlation, plus a separate representation-level diagnostic gap relative to feature references that is not closed by removing constraints.

The Discussion situates these findings for simulation use, states what was and was not validated without post-dose EEG, and lists failure modes specific to this architecture and label structure.

4. Discussion

What worked at the locked fold-0 checkpoint is external signature concordance: effect-vector correlations reached $r = 0.908, 0.851, \text{ and } 0.918$ on TUH, OSF, and P-ADIC with 10/10 secondary sign agreement. What did not generalize robustly is continuous similarity across training folds (five-fold means near 0.37 to 0.39), and an equal-weight ensemble ($r = 0.642/0.626/0.627$) does not clearly exceed the prior-only mean $r = 0.636$. What happened to diagnosis is attenuation, not recovery: CAUEEG latent probe ROC-AUC = 0.579 sits below theta/alpha 0.675 and raw 2185-D 0.699, and nested heads peak at out-of-fold ROC-AUC = 0.593. Computationally, these results are consistent with a literature-constrained response objective that can produce externally concordant response structure at a selected checkpoint while representation learning attenuates diagnostic information relative to explicit feature references. They do not establish empirical post-dose pharmacodynamics, fold-stable continuous fidelity, or external diagnostic utility.

Signature concordance should be read against EEG digital-twin work that targets diagnosis and prognosis rather than chemistry-conditioned response simulation. Digital Alzheimer’s Disease Diagnosis (DADD) twins invert EEG into neurodegeneration-linked parameters and report strong CSF and conversion prediction relative to standard EEG metrics [17, 18]. Related personalized cortical models likewise estimate synaptic and connectivity degeneration severity from EEG for early-stage stratification [19]. Those lines of work show that mechanism-informed EEG twins can carry clinical signal when diagnosis is the designed endpoint [20, 21]. Parallel EEG pharmacology and AD electrophysiology work shows that cholinergic intervention, slowing, and network disruption are detectable in spectral and connectivity EEG [14, 15, 10, 11, 31, 32, 33, 34]. Our twin sits in a different niche: ChemBERTa-conditioned simulation under explicit band and connectivity priors. The prior-only baseline (10/10 signs; mean $r = 0.636$) shows that directional agreement is partly recoverable from the literature constraint itself, so higher locked fold-0 continuous similarity should not be interpreted as independent discovery of the drug direction. Separately, a logistic domain classifier on the same 2185-D features separates training from TUH, OSF, and P-ADIC with pooled out-of-fold AUCs of 0.984, 1.000, and 1.000. Feature-family forensics show that this separability is distributed across PSD, band-power, coherence, and PLV families and persists under dimensional compression, rather than being concentrated in a very small number of individual dimensions. Those AUCs quantify representation-level distributional differences; they do not establish that domain shift caused the twin’s direction or magnitude results, and they are not evidence of domain adaptation [29].

The diagnostic null and encoding gap parallel a known dissociation in generative modeling. Variational autoencoders already separate reconstruction, disentanglement, and predictive utility in other domains [1, 2, 4], including transcriptomic drug-response VAEs where reconstruction quality need not improve response prediction [3]. What is specific here is the endpoint pair: external

signature-direction concordance versus external diagnosis from latent magnitude, with residual content at μ_{base} scored against feature references on the same subjects. That is a contribution of measurement design, not a claim of unprecedented generative-model behavior [29, 35].

Practically, the twin is best used for signature-constrained treatment-response simulation and hypothesis generation, not as a standalone external diagnostic classifier or as a post-dose pharmacodynamic validator. If diagnosis is required, calibrate against feature-level references such as theta/alpha (ROC-AUC = 0.675) or the full 2185-D vector (ROC-AUC = 0.699); performance near 0.58 to 0.59 on frozen μ_{base} is partial survival, not recovery of input clinical information.

Limitations are specific. There is no paired post-dose EEG ground truth in the external cohorts. Continuous direction fidelity depends on the locked fold-0 checkpoint, with high fold-to-fold variance and no clear ensemble increment over the prior. The training cohort ($N = 66$) is small relative to 2185-D inputs and multilayer CVAE capacity. Primary results use a single training seed family (seed 42). A ChemBERTa-versus-one-hot ablation was not run. Checkpoint selection follows a documented fold-0 rule matched to the unconstrained control; historical chronological lock is not independently proven. CAUEEG "Dementia" is a clinical spectrum with label heterogeneity. Domain shift is demonstrated as representation-level separability but not causally decomposed into acquisition or biological components. The α_c sweep is exploratory rather than a full nested CV of constraint weight.

A concrete next experiment is to train matched twins that keep, anneal, or remove constraints and add an auxiliary diagnostic head calibrated to a held-out feature reference, then repeat the three-cohort direction protocol plus CAUEEG encoding battery. The decisive question is whether latent probe AUC can move toward 0.675 to 0.699 without dropping fold-0 external correlations below the constrained operating range reported here.

The Conclusion restates what this study establishes and what remains out of scope.

5. Conclusion

This study shows that a literature-constrained EEG-drug CVAE can generate externally concordant response signatures at a locked fold-0 checkpoint (TUH/OSF/P-ADIC $r = 0.908/0.851/0.918$; secondary sign agreement 10/10), while continuous fidelity is sensitive to fold selection (five-fold means about 0.37 to 0.39) and an equal-weight ensemble ($r = 0.642/0.626/0.627$) does not clearly exceed the prior-only baseline ($r = 0.636$). Latent diagnostic information is substantially attenuated relative to explicit feature references (latent ROC-AUC = 0.579 versus theta/alpha 0.675 and 2185-D 0.699; best head OOF AUC = 0.593). The findings support signature-constrained treatment-response simulation and hypothesis generation, rather than empirical post-dose pharmacodynamic validation or standalone external diagnosis.

CRedit authorship contribution statement

Muhammad Haris: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review and editing, Visualization.
Syed Abdul Rehman Nasir: Software, Validation, Investigation, Writing - review and editing.

Muhammad Hamza Sajjad: Software, Investigation, Writing - review and editing. **Hikmat Ullah Khan:** Conceptualization, Methodology, Writing - review and editing, Supervision.

5.1. Ethics statement

This work is a secondary analysis of de-identified EEG datasets. No new human participants were recruited and no clinical interventions were performed. Training resting-state EEG features derive from the public OpenNeuro accession ds004504 (Miltiadous et al.). External cohorts were obtained under each provider’s academic data-use terms: the Temple University Hospital (TUH) EEG Corpus data use agreement; a public Open Science Framework (OSF) Eyes_closed AD/HC partition; the Dryad P-ADIC release (DOI 10.5061/dryad.8gtht76pw); and academic, non-commercial access to CAUEEG granted by the CAUEEG team (access contact: Min-jae Kim), with no redistribution of raw CAUEEG or TUH recordings. Analyses were conducted for academic research in the Department of Computer Science, University of Wah, under supervisor oversight. The study does not claim prospective clinical decision support.

5.2. Funding

This research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors. It was conducted as a Bachelor of Science (Artificial Intelligence) Final Year Project at the University of Wah.

Data availability

Training resting-state EEG features used in this project are from OpenNeuro accession ds004504 (Miltiadous et al. AD/HC resting EEG), processed to the feature arrays described in Methods. Temple University Hospital (TUH) clinical EEG is available to researchers under the TUH corpus data use agreement and cannot be redistributed by the authors. The OSF Eyes_closed AD/HC features used for the AD/HC external magnitude and direction checks were imported from a public OSF-hosted partition already present in the project archive. P-ADIC AD and control matrices are available from Dryad (DOI 10.5061/dryad.8gtht76pw) under the release terms of Shor et al. CAUEEG raw recordings were obtained by academic application to the CAUEEG team (access granted by Min-jae Kim; Kim et al., NeuroImage 2023) for non-commercial research use and must not be redistributed. Derived project artifacts that can be shared subject to those upstream terms include processed feature arrays generated in this repository (where redistribution is allowed), validation JSON summaries, and analysis code. Raw CAUEEG and TUH EDFs will not be redistributed by the authors. File-level paths are listed in Supplementary Material S1.

5.3. Code availability

Analysis and figure-generation code for this manuscript live in the project repository. The Python environment is the project virtualenv `eeg_twin` (Python 3.10); install from `requirements.txt` into that environment. Exact script names, configuration keys, checkpoint aliases, and regeneration commands are listed in Supplementary Material S1 and paper/CODE_AVAILABILITY.md. A public mirror (URL and commit hash) will be linked at acceptance; until then, the packaged Overleaf zip and the local repository are the reproducibility sources for the reported JSON and figures.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Appendix A. Reproducibility appendix

This appendix lists configuration keys, scripts, and artifact paths used to regenerate the manuscript numbers. It is not required to follow the scientific argument in the main text.

Appendix A.1. Model configuration

- Variant: `constrained_mlp` in `src/models/config.py` (`MODEL_VARIANTS['constrained_mlp']`).
- ChemBERTa input dim: `MODEL_CONFIG['drug_encoder']['input_dim'] = 384`.
- CVAE latent dim: `MODEL_CONFIG['cvae']['latent_dim'] = 128`.
- Hidden widths: EEG encoder [256, 512], drug encoder [256, 128], CVAE [512, 256]; `use_attention: False`.
- KL weight: `TRAINING_CONFIG['beta_kl'] = 0.01`.
- Constraint schedule: `warmup_epochs = 5`, `paper_constraint_weight = 0.1` in `run_constrained_training.py`.

Appendix A.2. Pharmacodynamic priors and loss helpers

- Priors store: `src/drugs/pharmacological_embedding.py` (`DRUG_PHARMACOLOGY[*]['eeg_effects']`).
- Prior loader: `get_eeg_effect_prior(drug_name)`.
- Band constraint: `compute_band_direction_constraint` in `src/models/losses.py`.
- Connectivity constraint: `compute_connectivity_constraint`.

- Band / connectivity index slices used by external direction scoring: BAND_SLICES in `src/tuh/tuh_validation.py` (delta [380:399], theta [399:418], alpha [418:437], beta [437:456], connectivity [475:1330]).
- Flatten order (`api/utils/feature_processor.py` `flatten_features`): PSD [0:380), band powers [380:475), then for each of five bands coherence upper-triangle (171) then PLV (171), so full connectivity is interleaved on [475:2185). Coherence and PLV are **not** contiguous pure blocks. The training connectivity constraint and direction-scoring window [475:1330) match `CONNECTIVITY_INDICES / BAND_SLICES['connectivity']`; that window mixes interleaved coh/PLV from early bands and must not be cited as an all-coherence block. Do not use `feature_block_ranges()['coherence_block']` as a pure-coherence map (it conflicts with interleaving).

Appendix A.3. Training data paths and checkpoint aliases

- Training arrays: `data/eeg_features/{AD,HC}*.npy` (PSD shapes (37, 19, 20) and (29, 19, 20)).
- Raw EEGLAB discovery: `EEGLoader.find_eeg_files` under `data/raw_eeg/{AD,HC}/`.
- Subject-level CV helper: `get_subject_level_kfold(n_splits = 5, random_seed = 42 from DATA_CONFIG)`.
- Constrained evaluation alias: `models/checkpoints_constrained/checkpoint_constrained.pt` (fold-0 best).
- Matched unconstrained alias: `models/checkpoints_unconstrained/checkpoint_unconstrained.pt` (from `checkpoints_5fold/fold_0_best.pt`).
- Simulation draws: `num_samples = 5` in `simulate_post_drug_eeg` for reported external runs.

Appendix A.4. External processing and validation artifacts

- TUH processing: `process_tuh_eeg.py` (`CROP_SECONDS = 320`, `TARGET_N_EPOCHS_FOR_PSD = 864`).
- CAUEEG processing: `src/tuh/process_caueeg_eeg.py`.
- Master report: `models/validation/complete_validation_report_v3.json`.
- Unconstrained battery: `models/validation/unconstrained_external_battery.json`.
- Prior-only baseline: `models/validation/prior_only_direction.json`.
- Fisher-z CIs for r : `models/validation/direction_r_fisher_ci.json`.

- α_c sweep: `models/validation/constraint_strength_sweep.json`.
- Latent probe: `models/validation/layer5e_caueeg_latent_probe_results.json`.
- Nested heads: `models/validation/layer5e_caueeg_classifier_head_results.json`.

Appendix A.5. Endpoint implementation notes

- Direction correlation key in JSON: `effect_magnitude_correlation`.
- Subject-bootstrap CIs: `models/validation/direction_r_subject_bootstrap_ci.json` (B=2000).
- Five-fold external direction: `models/validation/fold_external_direction_results.json`.
- Equal-weight fold ensemble: `models/validation/fold_ensemble/fold_ensemble_external_direction_results.json`.
- Ensemble subject bootstrap (B=2000, seed 42): `models/validation/fold_ensemble/bootstrap/ensemble_subject_bootstrap_B2000.json`.
- Prior vs ensemble incremental-value note: `models/validation/prior_vs_ensemble/` (paired Δr not supportable from stored prior-only artifacts).
- Domain-shift LR AUCs (2185-D): `models/validation/domain_shift/domain_shift_results.json`.
- Domain-shift forensics (feature families): `models/validation/domain_shift_forensics/domain_shift_forensic_summary.md`.
- Feature dimension map (interleaved connectivity): `models/validation/domain_shift_forensics/feature_dimension_mapping.md`.
- Raw 2185-D probe: `models/validation/layer5e_caueeg_raw2185_probe_results.json`.
- Magnitude score helper: `mean_drug_response`.
- Cohen's d helper: `_cohens_d` (sample variance, ddof = 1).
- Encoding probe scripts: `run_latent_probe_caueeg.py` (PROBE_DISEASE_LABEL = 0.0), `run_latent_classifier_head_caueeg.py`.
- Probe CV: `StratifiedKFold(n_splits=5, shuffle=True, random_state=42); N_PERM = 200`.

- Theta/alpha mean-fold AUC 0.675 vs pooled OOF AUC 0.673: `feature_probe_theta_alpha_dementia_vs_normal.cv.roc_auc.mean` vs `.cv.oof.roc_auc`.
- Raw 2185-D mean-fold AUC 0.699: `raw2185_logistic_dementia_vs_normal.cv.roc_auc.mean`.

Appendix A.6. Regeneration entry points

- `eeg_twin/Scripts/python.exe -m src.validation.run_prior_only_direction`
- `eeg_twin/Scripts/python.exe -m src.validation.compute_direction_r_cis`
- `eeg_twin/Scripts/python.exe -m src.validation.run_unconstrained_external_battery`
- `eeg_twin/Scripts/python.exe -m src.validation.run_alpha_c_sweep -eval-only`
- `eeg_twin/Scripts/python.exe src/figures/make_fig6_constraint_strength.py`
- `eeg_twin/Scripts/python.exe src/figures/make_graphical_abstract.py`

See also `paper/CODE_AVAILABILITY.md`.

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